

Noise-Resolved Physical Contrasts for Dexterous Interaction Models: A Pilot Study of Measurement and Actor-Panel Transfer

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Exploratory review copy. Rendered from the LaTeX draft using ReportLab. The source contains the publication equations; this copy displays equivalent plain-text equations. Both pilot gates failed. No policy or journal-readiness claim is made.

Abstract

Interaction models intended for dexterous control must distinguish the consequences of candidate actions, rather than merely predict observed motion. Before learning these differences, their physical measurement must also be resolved against repeated-process variation. We implement a cold-start replay diagnostic with one-step wrist translation interventions in an existing self-trained Inspire-hand simulation substrate. Four panels, each containing 96 environments across three airplane reference motions, are evaluated under two zero interventions and positive and negative 10-mm requested PD-target offsets. At five and ten control steps, the pooled position-response contrasts are 7.15 and 18.83 mm RMS, compared with 0.10 and 0.44 mm for zero-intervention repeats. However, vertical response is not consistently positive across panels, failing the predeclared joint physical screen. A separately documented vector-prediction screen then fits matched factual, differential-loss and no-pulse-input networks on one frozen actor's panels and evaluates on another's. Across three optimization seeds, differential-loss contrast errors are 10.11 and 26.99 mm, with no improvement over strong simple controls. These pilots separate measurable action effects from beneficial lift and from transferable learned effects. They motivate further work on contact-specific data coverage, while providing no evidence of manipulation-policy utility.

1. Introduction

Human hand-object data and simulated robotic rollouts offer substantial supervision for interaction dynamics. A predictive representation can nevertheless fail to improve a control decision. Prediction may capture common reference motion, while useful action selection depends on much smaller differences between feasible interventions. A second issue is experimental resolution: two closed-loop evaluations can have nearly identical aggregate success rates but disagree about which episodes succeed. Measurement quality, learning quality and task benefit therefore require separate checks.

The existing repository contains self-trained manipulation actors, object-centric motion models, cross-source geometry and intervention tooling. Its prior four-seed effect-rank validation does not support the joint positive action-alignment and policy-utility claim. A subsequent frozen-actor audit reports 34 versus 35 successes among 384 paired episodes, but 63 discordant success labels. These are inherited records, not newly validated results of this paper. They motivate measuring shorter physical responses before committing further resources to a controller.

Particle representations already support cross-embodiment world modeling and planning [he2025]. Value-aware model learning [farahmand2017] and action-discriminative world models [qiu2026] explicitly connect predictions to decisions. Controllability representations also have established foundations [rudolph2024]. We do not claim novelty for those general ideas. This pilot asks a narrower question: when a small action intervention is physically resolvable in a contact-rich substrate, does a simple contrast-supervised predictor transfer its effect to another actor's replay panels?

2. Physical-response measurement

For a recorded action sequence, let τ_i be the control step immediately after the first reference event where both the native hand-force and object-force proxies are active. This retrospective diagnostic schedule does not establish pairwise hand-object contact. For environment i , amplitude u in $\{-0.01, 0, 0.01\}$ modifies only wrist-z normalized action at τ_i . The native translation scale is one, giving a requested target offset of u meters. Actual wrist motion is governed by the

PD controller and contact physics; it is not asserted to equal the requested offset. All later actions follow the identical recorded sequence.

Define the response relative to each run's actual pre-intervention state as

$$R(i,H,u) = p(i,\tau+H,u) - p(i,\tau,u); D(i,H) = R(i,H,+) - R(i,H,-).$$

Two zero-intervention processes give the repeat contrast $N_{\{i,H\}} = R_{\{i,H\}}^{\{0a\}} - R_{\{i,H\}}^{\{0b\}}$. We report

$$S(H) = \sqrt{\text{mean}_i \|D(i,H)\|^2}; B(H) = \sqrt{\text{mean}_i \|N(i,H)\|^2}.$$

The descriptive ratio S_H/B_H is reported only when $B_H > 0$. A zero observed repeat difference does not establish zero process uncertainty or infinite measurement quality. It is encoded as an undefined ratio.

The DExplore simulation substrate [xu2025] uses self-trained actors in this study. Each process starts from the same archived cold simulator state, with physical-property, root/DOF, observation, recurrent-state and RNG checks inherited from the paired evaluator. No warm PhysX solver state is claimed to be restorable. Recorded per-step RNG schedules and normalized actions are replayed. Four panels combine actor training seeds 286/287 with evaluation seeds 288/289; each panel has 32 environments per motion. These 384 windows are not 384 independent motions or training seeds. The predeclared physical screen requires $S_H/B_H \geq 2$ at $H=5,10$, and positive mean vertical contrast in every panel at both horizons. This is an operational Probe gate, not a statistical confidence guarantee. The matrix has no failures or excluded runs.

3. Vector-prediction screen

After the physical screen fails its directional component, a documented decision retains that failure and permits a separate test of full vector-response prediction. Its reused test panels were already included in physical diagnostics; they are not a pristine Validation set. Optimization and normalization nevertheless exclude all actor-287 rows.

The fit/test sets contain 192 windows each, from actor 286/287 respectively. A window has four intervention arms. Inputs comprise four actual past/current 55-dimensional physical states and masks, followed by ten predetermined normalized 18-dimensional action vectors beginning at the intervention. Future observed physical states never enter the features. The action sequences are retrospective replay banks, not deployable future actor actions. Targets are three-dimensional position responses at five and ten steps.

All learned models use two 128-unit SiLU hidden layers and six outputs. Three initialization seeds (301-303) each train factual, differential and no-pulse models for exactly 1000 Adam updates, learning rate 10^{-3} , with matched 96-window mini-batches. No checkpoint is selected using test errors. Normalization and loss scales use fit data only. Factual training minimizes normalized response MSE. Differential training adds coefficient-one MSE on predicted plus-minus contrasts, scaled by fit contrast RMS. The no-pulse model substitutes the zero-arm plan, retaining actual observed history and the common replay plan but hiding the assigned pulse. Additional controls predict the global fit mean contrast and a privileged per-motion fit mean contrast.

The fixed learning gate requires, at both horizons, at least 10% lower seed-mean contrast RMSE than the factual model and global fit-mean control, and at least 75% agreement with the measured vertical contrast's sign. Sign agreement concerns the observed sign, not an assumption that every effect should be positive. Optimization seeds quantify variation of this small fitting procedure; they are not independent policy-training replications.

4. Results

Steps	Seconds	Pulse RMS (mm)	Repeat RMS (mm)	Ratio
1	0.033	1.914	0.000	undefined
2	0.067	2.572	0.084	30.75

Steps	Seconds	Pulse RMS (mm)	Repeat RMS (mm)	Ratio
5	0.167	7.150	0.101	70.44
10	0.333	18.827	0.444	42.36
20	0.667	75.463	6.081	12.41
30	1.000	105.792	9.638	10.98

Table 1. Physical contrast measurements, pooled over the four fixed panels. Requested PD target offsets differ by 20 mm between positive and negative arms.

Physical-resolution subgates pass, but vertical contrasts have mixed signs. At five steps, panel means are -0.025, 0.009, -0.108 and 0.845 mm; at ten steps they are -0.028, 0.005, -0.303 and 0.253 mm. The joint physical screen is therefore UNPROMISING. Large three-dimensional response is not evidence that lifting the wrist improves object lift. Pre-intervention plus-minus position mismatch is zero in three panels and 0.012 mm RMS in the remaining panel. This measures exposed state agreement and does not establish equality of hidden solver state.

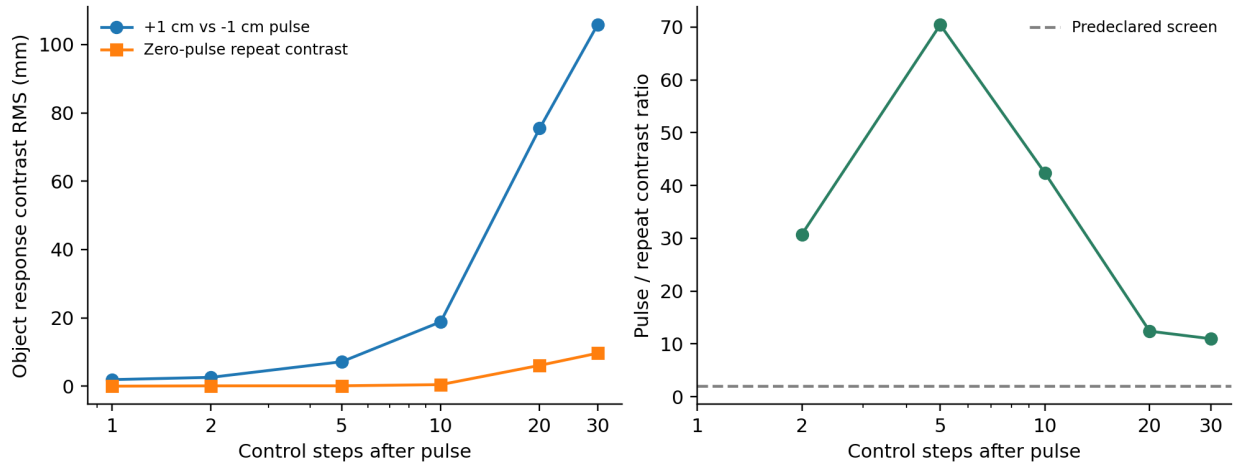


Figure 1. Physical pulse and repeat contrasts. Ratios summarize the observed repeat pair only; they are not calibrated signal-to-noise estimates or confidence bounds.

Model	Five steps (mm)	Ten steps (mm)
Factual	10.121	26.546
Differential	10.114	26.990
No-pulse input	9.922	26.456
Global fit mean	10.061	26.259
Per-motion fit mean	10.137	26.340

Table 2. Actor-287 contrast RMSE in millimeters. Learned rows average the three fixed optimization seeds. The motion control uses privileged motion identity.

Differential training changes mean error by less than 0.1% relative to factual training at five steps and increases it by approximately 1.7% at ten steps. It does not beat the global-mean baseline. Its measured vertical-sign agreement is 32.6% and 31.1%, below the fixed 75% gate. Every component of the learning screen fails, so this implementation is also UNPROMISING. The complete physical matrix took 481.9 seconds, and the nine GPU fits plus preprocessing and evaluation took 17.2 seconds. One RTX 3090 was used. Combined run artifacts occupied approximately 70 MB before the manuscript export.

5. Discussion and limitations

The physical and learning screens answer different questions. The pulse experiment shows differences larger than observed zero-replay contrasts in this fixed substrate. It does not show that a representation can infer those differences in new settings. The learned screen directly fails to establish such transfer for the tested small model. Neither observation identifies the cause of failure. Sparse contact-state coverage, distribution change between actors, feature insufficiency and optimization remain candidate explanations; this study does not select among them.

Contact proxies can activate because the hand and object independently contact the table. A future measurement protocol should validate actual interacting geometry or pairwise contact information. Shared cold conditions and replay actions also do not clone hidden contact solver state. Run order is fixed, with only one positive and one negative process per panel, limiting treatment/noise separation. No population-level confidence interval is claimed, and no inferential advantage is inferred from the nominal count of 384 repeated windows.

The studied object and reference motions are shared between fitting and evaluation. The action banks are diagnostic and involve only one wrist direction. There is no learned rollout controller, matched Cm-on/off policy training, real-world experiment, unseen-object test, or cross-hand test in these pilots. Strong geometric and model-based control baselines would be required before attributing benefit to a new interaction representation. The results cannot justify a journal-level generalization or utility claim. Any proposed future method must also be distinguished from established cross-embodiment, value-aware and action-bisimulation approaches.

6. Conclusion and next research decision

The tested short-horizon physical effects are measurable, but a monotone wrist-lift interpretation fails and the tested differential predictor does not transfer across the frozen actor panels. We retain both negative Probe decisions and stop this small-data model rather than increasing its updates or selecting another seed. The next consequential question is whether verified contact-conditioned coverage changes predictability on newly collected, independently frozen test data. Policy and hardware utility would remain separate required validations even after a positive predictive result.

Reproducibility

The study is isolated in agent/contact-response-cm. Physical execution is fixed to commit 466e80d; learning execution to 44f22ea. Run manifests record input/output hashes, commands, GPU UUID, process status and budgets. Original checkpoints, motion data and previous audits are read-only inputs. Raw compact physical traces, nine fitted networks and test predictions are retained. The authoring script exports tables directly from the recorded JSON results. The source repository's original working tree remains on agent/paired-evaluator.

References

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